A Hyperparameter-Free Attention Module Based on Feature Map Mathematical Calculation for Remote-Sensing Image Scene Classification

Qiao Wan, Zhifeng Xiao, *Member, IEEE*, Yue Yu, Zhenqi Liu, Kai Wang, and Deren Li

***Abstract*— Remote-sensing scene classification (RSSC) is crucial for remote-sensing image interpretation and has become a research hotspot in recent years. However, the high complexity of remote-sensing scenes causes most RSSC models to fail to accurately capture key objects, resulting in low classification accuracy. Meanwhile, it is intractable to effectively distinguish similar scenes, such as forest and meadow, whose semantic labels are mainly determined by wide-scale features. In addition, existing remote-sensing attention mechanisms are heuristic settings, which require expert knowledge and extensive experiments. To solve the above problems, a novel plug-and-play hyperparameter-free attention module (HFAM) based on feature map mathematical calculation is proposed in this work. HFAM uses statistical indicators to quantitatively characterize the fluctuations of feature maps that can accurately locate key features and distinguish different scenes, alleviating the problems of intraclass diversity and interclass similarity. Moreover, HFAM adaptively acquires attention weights by performing simple mathematical calculations on the feature maps, which solves the problem of difficult adjustment of hyperparameters. Our proposed HFAM can be expediently inserted into the existing ConvNet models without increasing the number of model's parameters. Extensive contrast experiments with several famous plug-and-play attention modules on three mainstream datasets reveal the superiority of our HFAM in accuracy, number of parameters, and calculation amount. Moreover, compared with state-of-the-art methods, it also demonstrated considerable competitiveness.**

***Index Terms*— Attention module, convolutional neural networks (CNNs), hyperparameter free, scene classification.**

I. Introduction

DUE to the rapid development of Earth observation technology, intelligent understanding and analysis of huge amounts of remote-sensing data subsequently become a growing requirement, which are widely applied in many practical applications such as environmental monitoring, land use surveys, land cover change detection, object detection, video understanding, semantic segmentation, and urban planning [1], [2], [3], [4], [5], [6], [7], [8], [9], [10]. Different from pixel-level classification such as land cover classification [11], [12] and remote-sensing scene classification (RSSC), which aims to represent high semantic information of remote-sensing scenes and classify remote-sensing images into corresponding labels set depending their different semantic meanings [13], [14], is a basic task in the field of remote-sensing research and provides a solid foundation for other applications such as object detection, object tracking, and image segmentation.

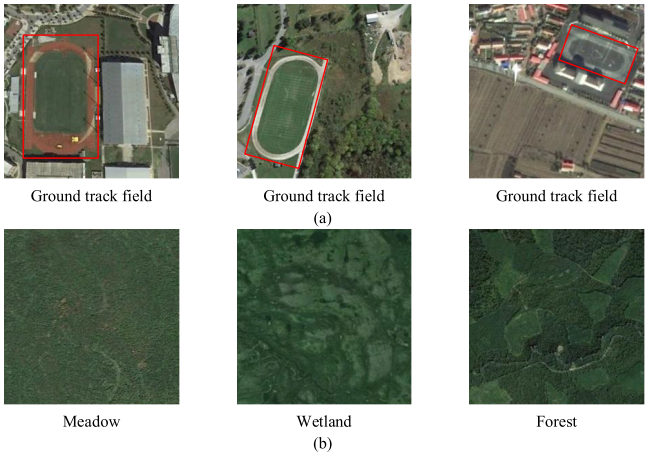


Fig. 1. Examples of remote-sensing images to demonstrate: (a) intraclass diversity and (b) interclass similarity. Red rectangles in (a) indicate the key target in the image that mainly determine semantic label.

However, due to the characteristics of multiscale, rotation and background complexity of high-resolution remote-sensing images, the difficulty of scene classification mainly come from two aspects: diversity for intraclass images and similarity for interclass images [15]. As shown in Fig. 1, intraclass diversity refers to the fact that images belonging to the same scene have large differences in visual characteristics. In the case of complex backgrounds, accurately identifying the important targets that determine the semantic label of images is the key for correct classification, such as the red rectangles in Fig. 1(a). On the other hand, interclass similarity is that images with different semantic labels exhibit similar visual characteristics. Fig. 1(b) is an example revealing the similarity between different kinds of images. It is a prospective solution

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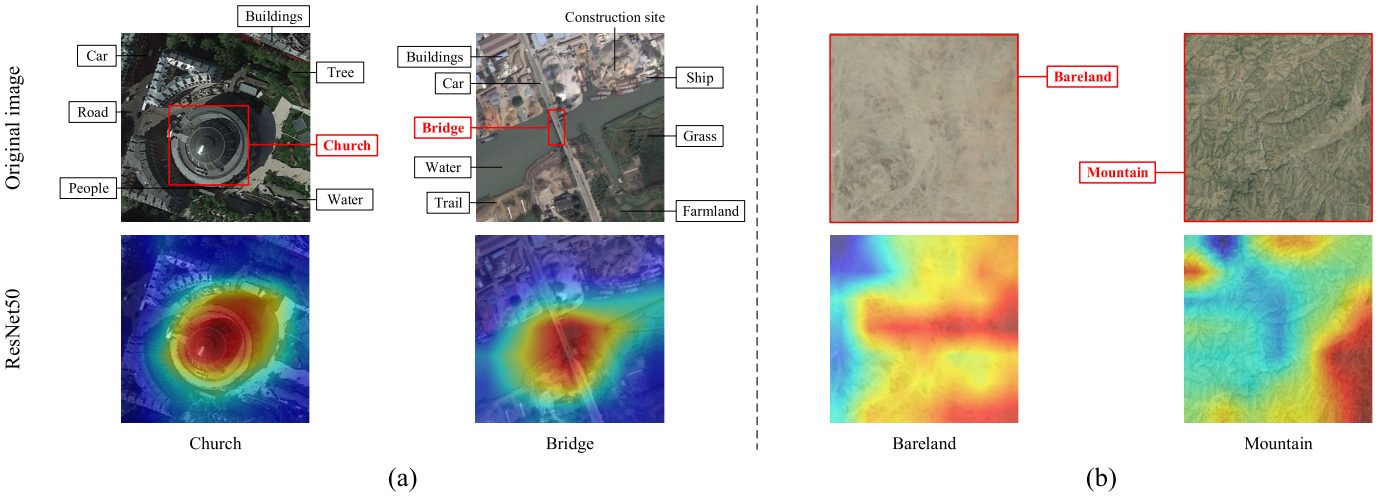


Fig. 2. Most two common types of remote-sensing scene categories. (a) Class A refers to the scenes whose semantic labels are determined by key objects or the relationships between them. (b) Class B represents the scenes whose categories depend on wide-scale features such as color, texture, shape, and spatial structure.

for interclass similarity to effectively distinguish these wide-scale similar visual features such as color, texture and shape.

Recently, benefiting from the huge success and outstanding performance of deep learning in computer vision (CV) field such as image classification [16], [17] and object detection [18], [19], deep learning-based methods features have become mainstream for the field of RSSC. Convolutional neural networks (CNNs) are considered to be the most successful and widely used deep learning technology for image processing, and thus various CNN-based methods have been designed for RSSC [20], [21], [22], [23].

Besides CNNs, a new deep learning paradigm of architecture called transformer has been proposed and gradually applied in the field of CV. The transformer proposed in [24] for natural language processing (NLP) tasks has become the mainstream in this field. The self-attention mechanism in transformer can perform global computations on long sequences of inputs and capture their dependencies. Inspired by the great success in NLP, the vision transformer (ViT) [25] was considered as the first work of the transformer-based model in CV, which directly feed the patches of the image into standard transformer encoder to learn the dependencies of different positions and extract contextual features of image. Recently, some valuable transformer-based works have been proposed and have achieved quite excellent results in the field of RSSC [26], [27], [28]. However, compared with CNN-based methods, the transformer-based methods have the following disadvantages. First, the CNN network has relatively smaller parameters of model and calculation amount than the transformer network, which makes CNN-based methods more suitable for some applications with limited computing resources [25]. Second, due to the lack of typical convolutional inductive biases, the transformer-based methods are more data-hungry than common CNNs and thus tend to perform worse than CNNs on small datasets [29], [30]. As we all know, the data magnitude of remote-sensing image datasets is far less than that of natural image datasets. The transformer-based methods used for remote-sensing are usually implemented through various powerful data augmentation techniques and customized complex networks special for remote-sensing images [26]. There is a risk of overfitting to train a huge model with a large number of parameters on a relatively small-scale dataset. Finally, the excellent ability of the transformer-based methods to obtain global relationships may lead to the loss of important local information of small-scale objects in remote-sensing images. Therefore, the CNN-based methods are still an irreplaceable and important research direction in the field of RSSC.

It is well known that a better CNN structure leads to better performances and therefore a great deal of research has focused on building satisfactory CNN structures. However, designing a brand new and powerful CNN network for RSSC requires comprehensive knowledge of remote-sensing images and CNN structures, which takes heroic efforts by experts to achieve. Thus, it is an important and effective directions to introduce attention mechanisms into CNN networks to emphasize the outputs of convolutional layers and acquire more discriminative features [31]. Naturally, various attention-based methods for RSSC have been proposed in recent years [9], [32], [33], [34], [35], [36].

Despite the promising progress these attention-based methods have made, they are still plagued by three problems. First, some attention-based methods are deeply integrated with specific backbone networks and even involve modification of cross entropy as loss function, lacking in portability and convenience. Therefore, in this article, we focus on developing a plug-and-play attention module that can be conveniently and flexibly utilized within most existing CNN models such as VggNet [16] and ResNet [17]. Second, most plug-and-play attention module are built by combinations of complex processes such as convolution and fully connected operations, which will inevitably increase the number of parameters and computation of baseline network. Specifically, to solve this problem, we propose a method based on feature map mathematical calculation and design several simple mathematical formulas guided by statistical

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indicators to directly obtain attention weights according to the characteristics of remote-sensing images. Lastly, the vast majority of attention-based modules have numerical preset hyperparameters, whose values are assigned before the learning process begins. Choosing suitable hyperparameters is usually critical for achieving satisfactory performance. However, it is quite difficult to select appropriate values for them and often requires lots of ablation experiments to determine. For example, the SE module [31] and all attention-based methods incorporating it have a hyperparameter of reduction ratio. Furthermore, for most attention modules, it is also necessary to predetermine the dimensions of the input and output of the attention module according to the different insertion parts in network, which is not convenient enough for implementation. Therefore, we mainly focus on designing a lightweight, hyperparameter-free, and plug-and-play attention module that can fully adaptively generate weights without increasing the number of parameters of baseline network and with little computational cost and can be easily embedded into existing CNN networks.

Consequently, we design a lightweight plug-and-play attention module named hyperparameter-free attention module (HFAM) to obtain attention weights based on feature map mathematical calculation, which is tailored for remote-sensing image scene recognition. It does not increase the number of parameters to baseline networks and achieves significant improvements without any data augmentation technology. Furthermore, compared with state-of-the-art attention-based methods for RSSC, it is quite competitive especially in terms of speed, number of parameters and hyperparameter free. In summary, our main contributions are as follows.

1) We design a new attention module for scene classification based on the characteristics of remote-sensing images. Our proposed HFAM can accurately locate key features and thus solve the problem of diversity for intraclass to a certain extent. Moreover, it uses statistical indicators to quantitatively characterize the fluctuation of feature maps, which can distinguish different scenes and solve the problem of similarity for interclass.

2) We simplify the complexity to generate attention weights by neat mathematical calculations on feature maps. Compared with existing attention modules, our approach heuristically and adaptively obtains attention weights, which solves the problem of difficult adjustment of hyperparameters, without increasing the amount of model parameters. It can be easily utilized by simple mathematical calculation on feature maps and expediently inserted into existing CNNs, which does not increase number of parameters.

3) We conducted extensive contrast experiments with several famous plug-and-play attention modules on three mainstream datasets including UC Merced (UCM) land use dataset [37], Aerial Image Dataset (AID) [13], and NWPU-RESISC45 (NWPU) [14] to verify the effectiveness of our HFAM and experimental results demonstrate that it achieved excellent performance in accuracy, number of parameters and calculation amount. Furthermore, compare with other state-of-the-art methods in RSSC, it also achieves competitive classification results.

II. Related Work

A. Remote-Sensing Image Scene Classification

Over the past decades, numerous methods have been proposed to improve the performance of scene classification in remote-sensing image. In general, they can be divided into three categories according to different feature representations they used: low-level-feature-based methods, middle-level-feature-based methods, and high-level-feature-based methods. Early scene classification methods for remote-sensing image were designed on the basis of low-level features such as color histogram (CH) [38], local binary pattern (LBP) [39], and scale-invariant feature transform (SIFT) [40]. Middle-level-feature-based methods attempt to extract feature representation of scene by computing local descriptors and then encoding them. The bag of visual words (BoVW) is a popular middle-level-feature-based method because of its simplicity [37], [41]. However, these traditional methods can seldom meet requirements in real applications. The geometrical structures and spatial patterns of remote-sensing images are very complex; therefore, the efficiency and performance of low-level-feature-based methods and middle-level-feature-based methods have no advantages, especially for large-scale datasets.

In recent years, benefiting from the huge success and outstanding performance of deep learning in CV field such as image classification [16], [17] and object detection [18], [19], deep learning-based methods extracting high-level features have become mainstream for the field of RSSC. CNNs are considered to be the most successful and widely used deep learning technology for image processing, and thus various CNN-based methods have been designed for RSSC. Zhang et al. [20] proposed a gradient-boosting random convolutional network (GBRCN) combining different deep CNNs to improve RSSC performance. Liu et al. [21] proposed a multiscale CNN model to extract scale-invariant features to eliminate the scale variation of the objects in remote-sensing image. Cheng et al. [22] employed the metric learning in feature learning and proposed an effective model for scene representation. Lu et al. [23] proposed a feature aggregation CNN model combining a supervised convolutional features encoding module and a progressive aggregation strategy for remote-sensing image scene classification tasks.

B. Attention-Based CNN Methods for Remote-Sensing Scene Classification

Attention mechanism, which improves the expressiveness of features by emphasizing crucial information and suppressing unnecessary background, has been introduced into various visual tasks and achieves excellent performance. The development of attention-based methods for RSSC is spontaneously inspired by the great success of attention modules. A large number of attention-based methods are proposed for remote-sensing image scene classification. Zhao et al. [32] directly added the CBAM [42] into residual dense network to improve the performance of RSSC. Yu et al. [33] proposed an improved channel attention scheme based on SE [31] is designed to enhance the features from different layers and

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then different enhanced features are bilinearly pooled and fused for classification. ARCNet is a recurrent neural network combined with an attention module, which can adaptively select important regions and obtain powerful predictions to correctly classify remote-sensing images [43]. CAD is a channel-attention-based network combined with DenseNet [44] and SE [31] which can achieve impressive performance for RSSC [34].

However, the above attention-based methods are directly utilized, modified, or improved on the basis of existing attention modules to apply in the field of remote-sensing image scene classification, while they are designed for natural images without considering the characteristics of remote-sensing images. Moreover, from the perspective of implementation, these methods are implemented by embedding the attention module into a specific network structure, which is difficult to extend to other existing CNN networks. Therefore, this article is dedicated to developing a plug-and-play attention module with good portability.

C. Plug-and-Play Attention Modules

Existing mainstream plug-and-play attention modules obtain weights by various operations in the channel domain, spatial domain, or their combinations. As one of the most representative channel attention modules, SE obtains the channel weights by a spatial average pooling operation and two fully connected layers, which can encourage the network to capture task-relevant features and suppress irrelevant background features. On the other hand, some studies are devoted to designing attention modules in spatial dimension. Nonlocal block [45] is a famous spatial attention module to generate the importance of different regions by a series of complex and computationally expensive operations to capture long-range dependencies. It can be found that these different attention modules are conducted on either channel domain or spatial domain.

Furthermore, some researchers have made attempts to study mixed attention mechanism that assigns exclusive weights to each neuron. A residual attention network is proposed to generate mixed weights by utilizing the encoder and decoder framework [46]. However, extremely poor expansibility is the fatal flaw of this method. It designs different subnetworks between lower layers and higher layers of a ResNet and is difficult to extend other networks. CBAM [42] is another attention module invented to acquire mixed weights by generating channel domain and spatial domain weighs, respectively, and then combining them. Complicated operations in CBAM make it too long to compute. An excellent mixed attention module should have the characteristics of operation easily and computation simply. Moreover, the lightweight property and extendibility of attention modules are also significant. SimAM [47] is a mixed attention module by introducing the concept of spatial suppression in visual neuroscience, which optimizes an energy function to determine the importance of each neuron and finally obtains mixed weights. However, this simple and effective module is designed for natural images and does not take into account the characteristics of remote-sensing images. Besides, it still has a hyperparameter. It is a well-known fact that the choice of hyperparameter is intractable and often requires a lot of ablation experiments to determine.

In the field of RSSC, some plug-and-play attention modules have been proposed for improving accuracy of backbone networks. EAM is an enhanced attention module improved on the basis of CBAM for remote-sensing image classification [35]. MLFC-net is a multilevel semantic feature clustering attention model implemented by a channel-wise attention mechanism and an ensemble structure for RSSC [36]. Nevertheless, they are designed by combinations of convolutional, pooling, and fully connected layers, which greatly increase the number of parameters and complexity of network. Meanwhile, they all involve hyperparameters and require extensive ablation experiments to determine their value.

Different from these previous plug-and-play attention methods, we aim to design a simple and efficient plug-and-play attention module without hyperparameters according to the characteristics of remote-sensing images, which can automatically and adaptively generate mixed attention weights based on feature map mathematical calculation and do not increase the number of parameters of model.

III. Motivation

In this article, we focus on two common remote-sensing scene categories: semantic labels are determined by key objects or the relationships between them (Class A) and scene classes are mainly determined by wide-scale image features such as color, texture, shape, and spatial structure (Class B). Then, we make the following conjectures for these two different scenes. For the former, it is critical to accurately identify and locate the key and related objects from complicated backgrounds, and thus it is necessary to focus on local important areas to improve the accuracy of classification. For the latter, their categories are mainly determined by wide-scale features such as color, texture, or spatial structure of the whole image. Different from the former, recognizing similar but different wide-scale features present in global scope is the key to correctly identify them. Fig. 2 shows some examples belonging to these two different types and their corresponding gradient-weight class activation mapping (Grad-CAM) [48] generated by ResNet50 [11], visually revealing the differences between them. The red boxes indicate the region of interest, which focus on the local key area in the image for Class A and global scope of the whole image for Class B. It can be observed that the corresponding heatmaps of different categories also confirm our conjectures: heatmaps generated by images of Class A such as bridge and church tend to focus on important local regions to identify the categories of scenes while heatmaps generated by images of Class B such as bareland and mountain prefer to pay attention to global scope to capture wide-scale feature of images.

Based on this, we make two assumptions for these two different types of remote-sensing scenes. One is that key objects are different from the background and their importance

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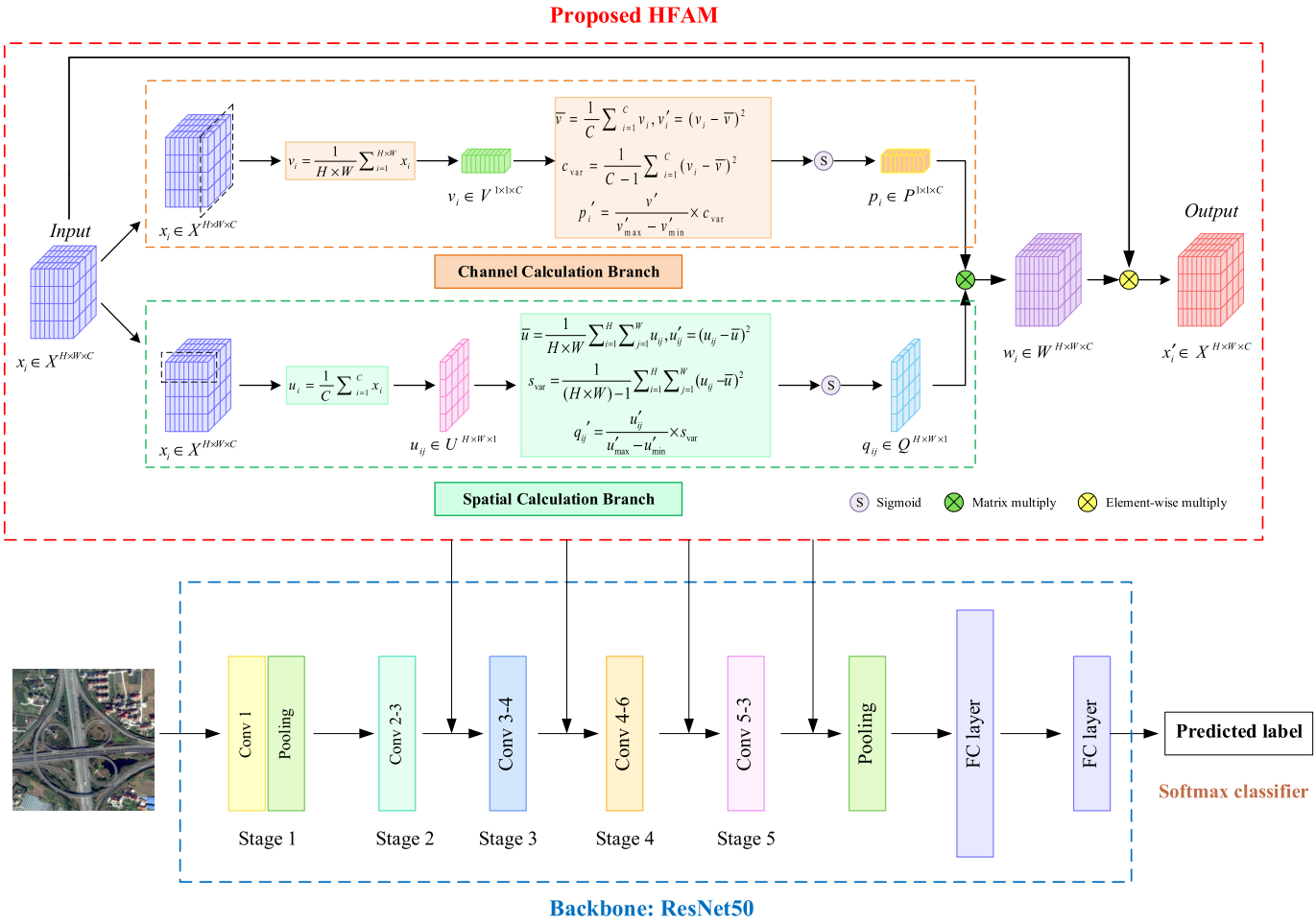


Fig. 3. Schema of our proposed HFAM module inserted into ResNet50. Our proposed HFAM computes channel attention weights and spatial attention weights in parallel by CCB and SCB and ultimately generate mix attention weights by matrix multiplication. HFAM adaptively obtains attention weights by simple and neat computations on feature maps without any hyperparameter.

can be measured by their difference from the mean value of the feature map. In this assumption, the mean value of the feature maps is taken as background information and the difference between each pixel and the mean value is used to measure the salience of this pixel. The other one is that wide-scale features can be recognized by the value range of the difference between each pixel on the feature map and the mean value and scenes with similar large-scale features can be distinguished by the variance of the difference. For scenes whose semantic labels are determined by a large range of texture features, such as forests and grasslands, the range of their feature maps from the mean value is quite different from scenes like airport and church. Furthermore, this value between them is also different in distribution and naturally similar texture features can be distinguished by the combination of it and its variance.

Based on the two assumptions above, we devise some simple formulas for calculating spatial weights and channel weights, respectively. Then, we multiply them to get ultimate mixed attention weights. Therefore, we design a novel attention module called HFAM to implement these operations. Furthermore, due to the multiscale characteristics of remote-sensing images, the key and related important objects in Class A are also multiscale variations. Inspired by the excellent performance of multiscale attention networks in Land Cover Change Detection with remote-sensing images [49], we insert the proposed HFAM module into different positions of the backbone CNN model to make it have multiscale characteristics and thus improve the recognition of those multiscale objects in remote-sensing images.

In summary, our proposed HFAM is a lightweight plug-and-play attention module tailored for RSSC, which can self-adaptively obtain attention weights by simple mathematical calculation on feature maps. For the above two common types of remote-sensing scenes, it computes the difference between each pixel and the mean value to measure the significance to accurately locate the key features in Class A and meanwhile uses statistical indicators to quantitatively characterize the fluctuations of feature maps to distinguish different scene in Class B. Compared with other plug-and-play attention modules, our HFAM has no hyperparameters and does not increase the number of parameters of the backbone networks.

IV. Methods

A. Overview

The proposed HFAM is completely implemented by simple mathematical calculations, without convolution and fully connection operations. It includes two parts: channel

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calculation branch (CCB) and spatial calculation branch (SCB). They parallelly compute channel and spatial attention weights and then mix them by matrix multiplication, so that the ultimate feature maps are more comprehensive and expressive.

In order to verify the effectiveness and generalizability of our proposed HFAM, three widely used CNN models including Vgg16 [16], ResNet50, and ResNet101 [17] are, respectively, employed as backbone. For ResNet50 and ResNet101, HFAM is executed after stage 2, stage 3, stage 4, and stage 5, respectively. For Vgg16, they are also embedded into multiple positions: after the fourth, seventh, tenth, and last convolutional layer separately. Fig. 3 detailedly shows the overall architectures integrated into ResNet50.

B. Channel Calculation Branch

Given the 2-D feature map $X \in R^{H\times W\times C}$ be the input feature, where H, W and C represent the height, width, and channel, respectively. Generally speaking, for a 2-D feature map, one dimension is the spatial dimension characterized by height and width and the other is the channel dimension. Channel-based attention mechanism treats each channel as a different feature detector and pays corresponding attention to different channels according to their importance, thereby enhancing or suppressing different channels. According to our assumption proposed in Section III, we use the fluctuation of each channel relative to the average of all channels as a measure of salience and design a series of simple mathematical formulas to quantitatively characterize this salience as channel attention.

First, we compute the mean values of the input feature map X on the spatial dimension to generate a vector $V \in R^{1\times 1\times C}$. We can write it as $V = [v_1, v_2, \ldots, v_C]$ and then compute the mean value $\bar{v}$ and variance c_{var} by

$$\bar{v} = \frac{1}{C}\sum_{i=1}^{C} v_i \tag{1}$$

$$c_{\text{var}} = \frac{1}{C-1}\sum_{i=1}^{C} (v_i - \bar{v})^2. \tag{2}$$

Let $V' \in R^{1\times 1\times C}$ be a vector and v'_i is an element of it. Here $v'_i = (v_i - \bar{v})^2$. We use $v'_{\max}$ and $v'_{\min}$ to represent the maximum and minimum values in V', respectively. Then, $P' = [p'_1, p'_2, \ldots, p'_C]$ can be acquired by the following equations:

$$p'_i = \frac{v'_i}{v'_{\max} - v'_{\min}} \times c_{\text{var}} \tag{3}$$

where $i = [1, 2, \ldots, C]$. Subsequently, we use the sigmoid function to activate it and acquire the feature weight of each channel $P = [p_1, p_2, \ldots, p_C]$.

Fig. 4 intuitively shows an example of employing our proposed mathematical method to calculate channel attention. We can get a vector $V \in R^{1\times 1\times C}$ by computing the mean values of the input feature map X on the spatial dimension. After a series of mathematical calculations, the final channel attention is applied to the original vector V. We can observe that channels with values close to the mean are suppressed, while important channels whose values deviate from the mean are emphasized.

C. Spatial Calculation Branch

From the perspective of spatial dimension, different regions of the image are not equally important for the task. Different from channel attention, spatial attention aims to force the network to focus on task-relevant important regions. We also designed corresponding formulas for spatial dimensions to perform similar operations to obtain spatial attention weights.

Analogously, we compute the mean of the input feature map $X \in R^{H\times W\times C}$ on the channel dimension to receive a matrix $U \in R^{H\times W\times 1}$ and u_{ij} is an element of U. Similarly, we compute the mean value $\bar{u}$ and variance s_{var} by

$$\bar{u} = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W} u_{ij} \tag{4}$$

$$s_{\text{var}} = \frac{1}{(H\times W)-1}\sum_{i=1}^{H}\sum_{j=1}^{W} (u_{ij} - \bar{u})^2. \tag{5}$$

Let $U' \in R^{H\times W\times 1}$ be a matrix. u'_{ij} is an element of it and is obtained by $u'_{ij} = (u_{ij} - \bar{u})^2$. Similarly, we use $u'_{\max}$ and $u'_{\min}$ to represent the maximum and minimum values in U', respectively. Let $Q' \in R^{H\times W\times 1}$ represent a matrix, and q'_{ij} is an element of it. Then, q'_{ij} can be computed by the following equations:

$$q'_{ij} = \frac{u'_{ij}}{u'_{\max} - u'_{\min}} \times s_{\text{var}} \tag{6}$$

where $i = [1, 2, \ldots, H]$ and $j = [1, 2, \ldots, W]$. After the sigmoid activation function, we obtain the spatial weights $Q \in R^{H\times W\times 1}$.

As shown in Fig. 5, we get spatial weights $Q \in R^{H\times W\times 1}$ after some simple mathematical calculations. By comparing the spatial feature map before and after applying spatial attention, it is obvious that regions deviating from the mean are emphasized and other regions are suppressed to varying degrees.

D. Combination of CCB and SCB

For an input feature map $X \in R^{H\times W\times C}$, we employ CCB and SCB in parallel to compute channel attention weights and spatial attention weights, respectively. Subsequently, we multiply the spatial weights and channel weights to obtain the ultimate mixed weights. Finally, the original feature map is multiplied by element-wise product operation with this final attention weights to acquire the eventual refined feature. The calculation process can be demonstrated in Fig. 4, and the final feature map $X' \in R^{H\times W\times C}$ is generalized by the following equation:

$$X' = (P \times Q) \cdot X \tag{7}$$

where $\times$ is matrix multiply and $\cdot$ is element-wise multiply. P and Q represent channel attention and spatial attention, respectively.

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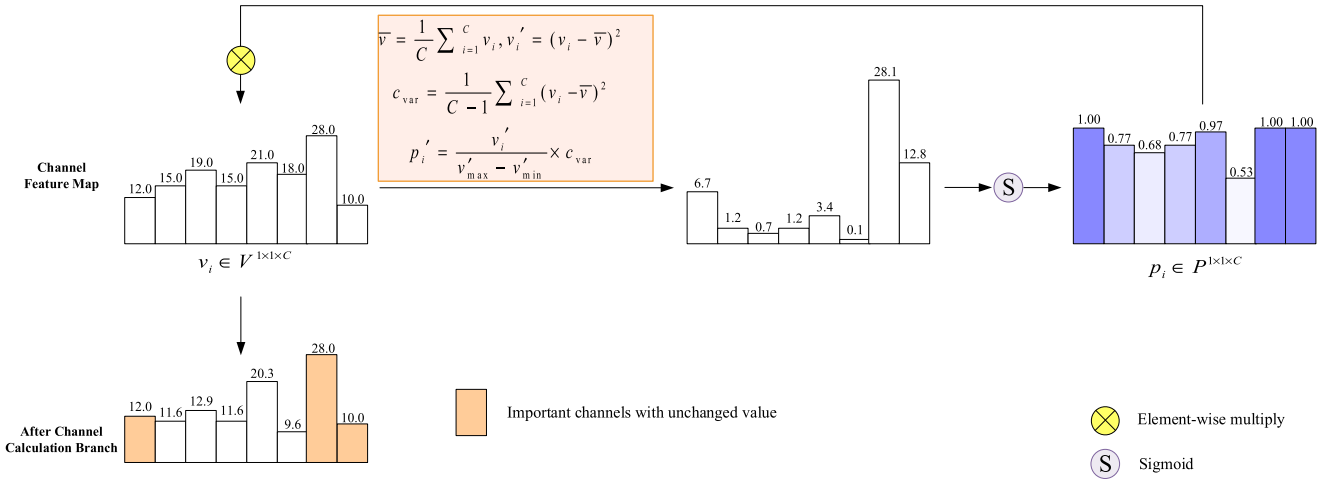


Fig. 4. Example to intuitively show the effect of CCB. It can be observed that channels with values close to the mean (17.3) are suppressed, while important channels whose values deviate from the mean are emphasized.

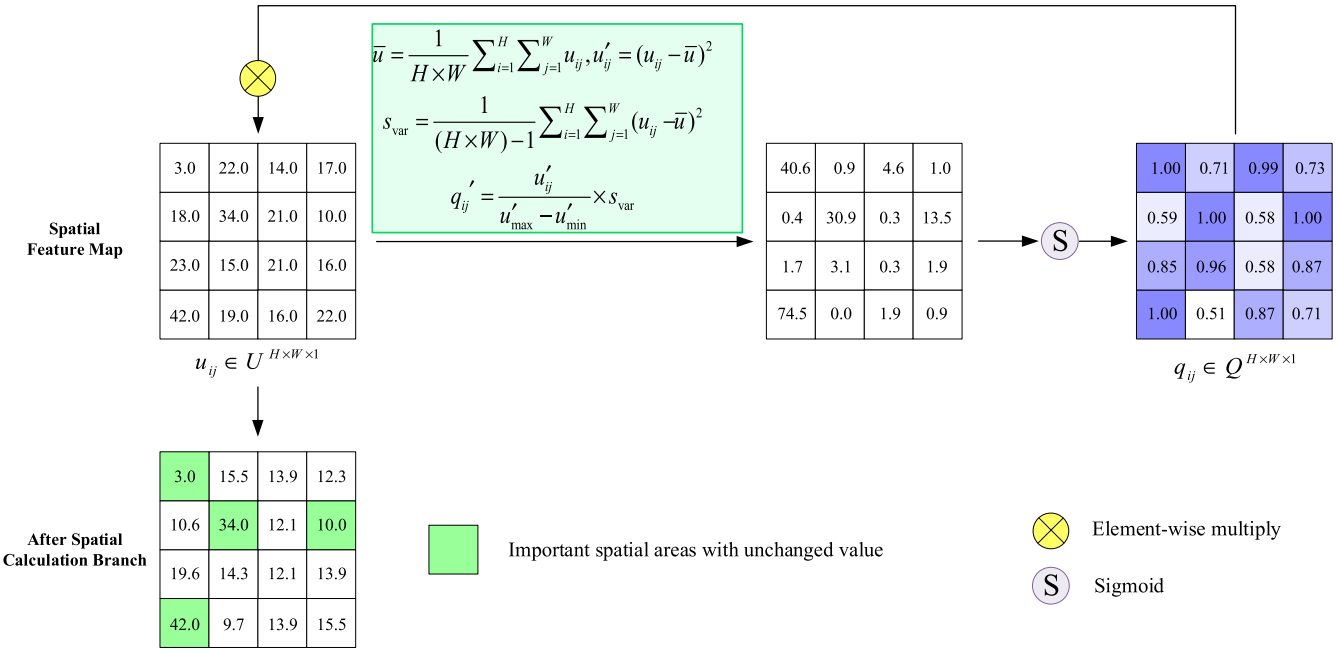


Fig. 5. Example to intuitively show the effect of SCB. Obviously, regions close to the mean value (19.6) in feature map are weakened while regions with a large difference from the mean are enhanced.

V. Experiments

In this section, we implement a mass of experiments on three widely used remote-sensing scene recognition datasets including the UCM land use dataset [37], AID [13], and NWPU [14], respectively, and compare our proposed HFAM with SE [31], CBAM [42], and SimAM [47] by inserting them into three existing classic CNN models. We reimplement all modules with consistent settings on the same workstation to get an impartial comparison.

A. Datasets

Experiments evaluating the performance of our attention module are conducted on three public and challenging remote-sensing scene datasets: UCM land use dataset [37], AID [13], and NWPU [14].

1) UCM Dataset: UCM land use dataset contains 21 classes of different remote-sensing scenes and each class has 100 samples with a size of 256 × 256 pixels in the red–green–blue color space. These 21 types of scenes are as follows: agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium density residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis courts. UCM dataset is widely used for remote-sensing image scene classification. In spite of the small scale, it is still challenging due to the high interclass similarity of some categories such as dense residential, medium residential, and sparse residential.

2) AID Dataset: AID is a large-scale dataset for aerial scene classification consisting of 10 000 images divided into 30 scene classes. There are 220–420 images with a size of

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TABLE I
DETAILS OF THE THREE DATASETS AND TRAINING RATIOS USED IN THIS ARTICLE

Dataset	Class	Image number	Images per class	Training ratio
UCM	21	2100	100	50%, 80%
AID	30	10000	220-400	20%, 50%
NWPU	45	31500	700	10%

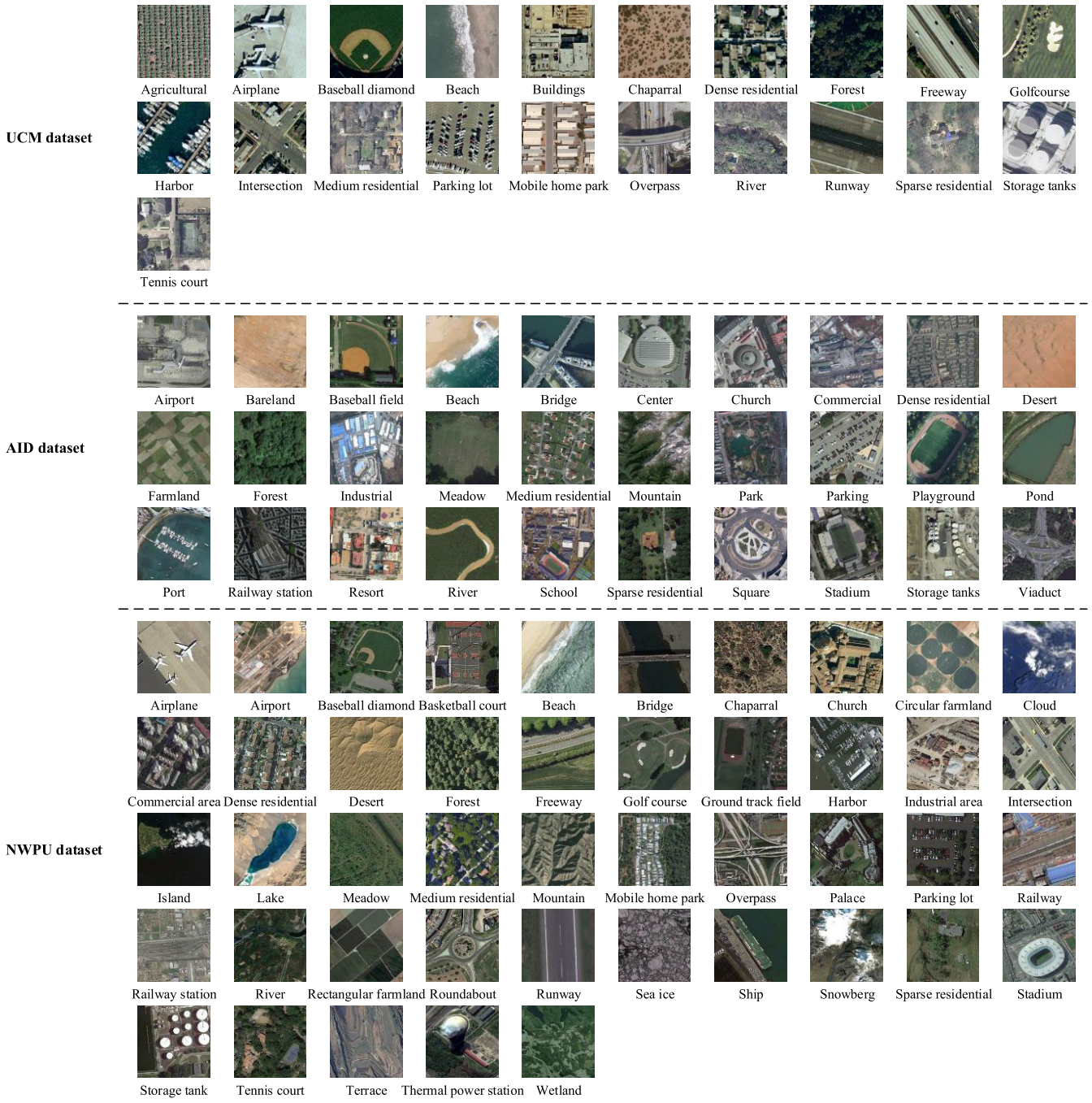


Fig. 6. Some samples from the three used datasets. Each image with different sizes from three datasets is scaled down to the same size in this figure for viewing convenience.

600 × 600 pixels in each class. These 30 scene classes include airport, bare land, baseball field, beach, bridge, center, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, park, parking, playground, pond, port, railway station, resort, river, school, sparse residential, square, stadium, storage tanks, and viaduct. Compared with the UCM dataset, the classification of the AID dataset is more challenging.

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TABLE II
COMPARISON BETWEEN DIFFERENT ATTENTION MODULES FOR DIFFERENT NETWORKS ON UCM DATASET

Model	OA on different training ratio (%)		Parameters (M)	Hyperparameter	FLOPs (G)
	50%	80%			
Vgg16	95.77 ± 0.25	97.81 ± 0.20	138.3870	×	248.3916
+ SE	95.89 ± 0.39	98.10 ± 0.17	+ 0.0758	√	+ 0.0466
+ CBAM	95.86 ± 0.37	98.19 ± 0.43	+ 0.0762	√	+ 0.0737
+ SimAM	96.00 ± 0.41	98.76 ± 0.32	+ 0	√	+ 0
+ HFAM	**96.42 ± 0.20**	**98.24 ± 0.36**	+ 0	×	+ 0.0197
ResNet50	97.06 ± 0.22	98.33 ± 0.34	25.5781	×	66.1386
+ SE	97.18 ± 0.29	**98.62 ± 0.31**	+ 0.6963	√	+ 0.0353
+ CBAM	96.90 ± 0.22	98.14 ± 0.39	+ 0.6967	√	+ 0.0530
+ SimAM	97.35 ± 0.14	98.43 ± 0.32	+ 0	√	+ 0
+ HFAM	**97.64 ± 0.18**	**98.62 ± 0.11**	+ 0	×	+ 0.0242
ResNet101	96.74 ± 0.24	98.38 ± 0.39	44.2755	×	125.8617
+ SE	97.31 ± 0.37	98.14 ± 0.52	+ 0.6963	√	+ 0.0353
+ CBAM	97.05 ± 0.32	98.62 ± 0.39	+ 0.6967	√	+ 0.0530
+ SimAM	97.43 ± 0.22	98.48 ± 0.21	+0	√	+ 0
+ HFAM	**97.46 ± 0.31**	**98.67 ± 0.21**	+0	×	+ 0.0241

TABLE III
COMPARISON BETWEEN DIFFERENT ATTENTION MODULES FOR DIFFERENT NETWORKS ON THE AID DATASET

Model	OA on different training ratio (%)		Parameters (M)	Hyperparameter	FLOPs (G)
	20%	50%			
Vgg16	91.23 ± 0.10	95.38 ± 0.19	138.3960	×	248.3918
+ SE	91.98 ± 0.31	95.45 ± 0.23	+ 0.0758	√	+ 0.0465
+ CBAM	91.91 ± 0.35	95.53 ± 0.07	+ 0.0762	√	+ 0.0736
+ SimAM	91.07 ± 0.21	94.99 ± 0.33	+ 0	√	+ 0
+ HFAM	**92.06 ± 0.16**	**95.78 ± 0.21**	+ 0	×	+ 0.0196
ResNet50	92.16 ± 0.18	95.51 ± 0.15	25.5781	×	66.1388
+ SE	92.77 ± 0.18	95.84 ± 0.22	+ 0.6963	√	+ 0.0352
+ CBAM	92.29 ± 0.15	95.38 ± 0.16	+ 0.6967	√	+ 0.0529
+ SimAM	92.23 ± 0.35	95.43 ± 0.17	+ 0	√	+ 0
+ HFAM	**93.11 ± 0.20**	**95.86 ± 0.15**	+ 0	×	+ 0.0241
ResNet101	92.15 ± 0.23	95.70 ± 0.13	44.5792	×	125.8618
+ SE	93.30 ± 0.13	**95.94 ± 0.14**	+ 0.6963	√	+ 0.0353
+ CBAM	92.34 ± 0.21	95.54 ± 0.18	+ 0.6967	√	+ 0.0530
+ SimAM	92.54 ± 0.20	95.54 ± 0.15	+ 0	√	+ 0
+ HFAM	**93.53 ± 0.21**	**95.94 ± 0.10**	+ 0	×	+ 0.0242

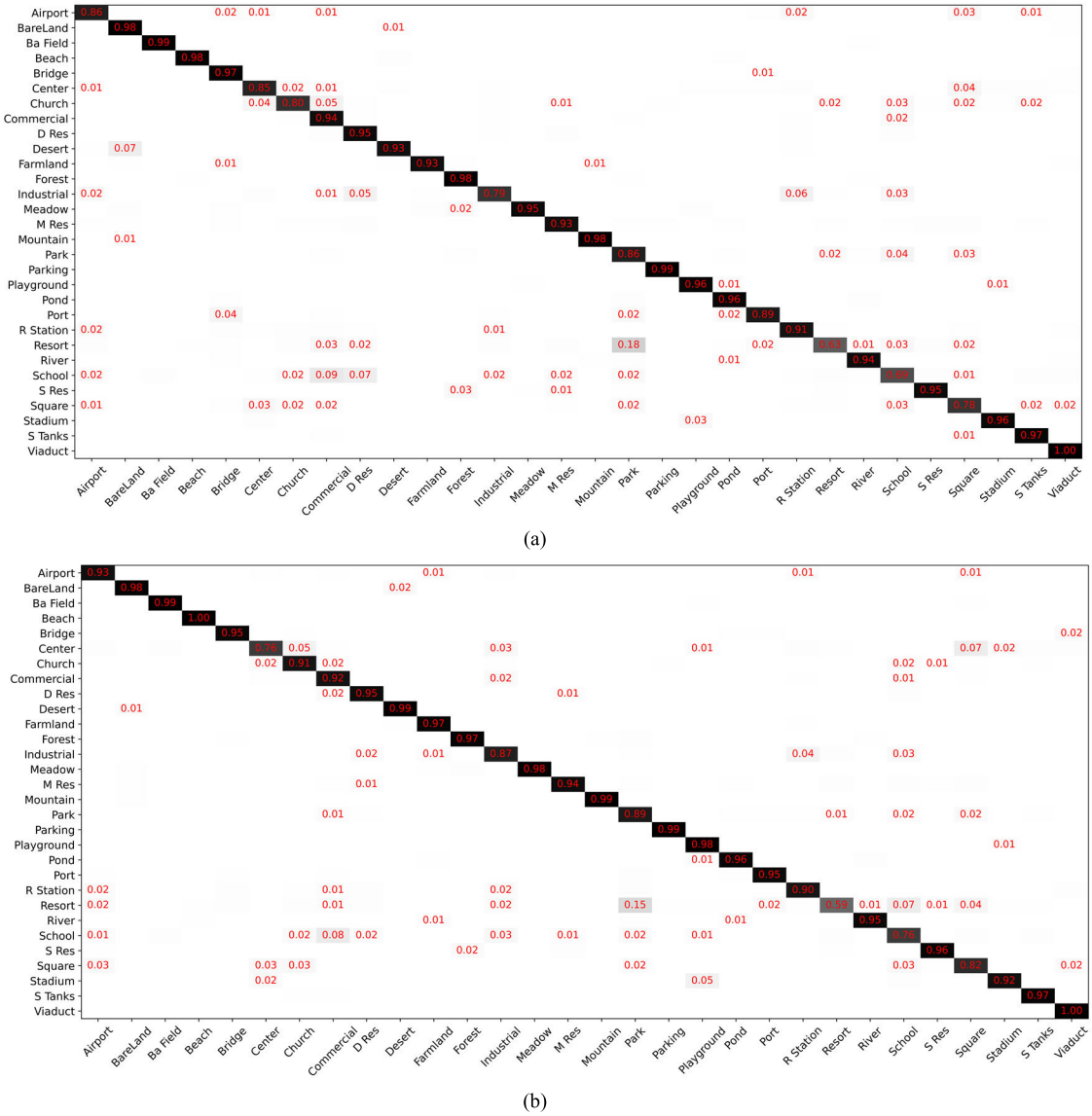
3) NWPU Dataset: NWPU is another public large-scale dataset for scene classification in remote-sensing images. This dataset has 31 500 images with a size of 256 × 256 pixels, which are divided into 45 classes including airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snowberg, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, and wetland. Big image variations, high within-class diversity, and between-class similarity make it more challenging.

The splitting strategy of the three datasets follows the conventional practice in the field of RSSC. Specifically speaking, UCM dataset is randomly split into 50% and 80% as the training set and 50% and 20% as the testing set, respectively. For the AID dataset, 20% and 50% of the images are randomly selected as the training samples, while the remaining 80% and 50% of images are used in testing. In the NWPU dataset, each category of images is randomly divided into 10% and 90% as training and testing sets. The details and the training ratio of the three datasets are shown in Table I. Additionally, some samples of the same or analogous categories in the three datasets are demonstrated in Fig. 6.

B. Experimental Setting

We conduct experiments on three baseline networks, respectively: Vgg16 [16], ResNet50, and ResNet101 [17]. For a fair comparison, we insert different attention modules including SE, CBAM, SimAM, and our proposed HFAM into the same locations (as illustrated in Section IV) of baseline models, respectively. In addition, it is a common strategy to use a pretrained model on ImageNet [50] to

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(a)

(b)

Fig. 7. Comparison of CM generated by baseline (ResNet50) and the network added our proposed attention module on AID dataset with 20% training ratio: (a) ResNet50 and (b) ResNet50 + HFAM.

fine-tune remote-sensing image dataset. Therefore, we take this measure to get better performance for all experiments. It is worth noting that we do not adopt any data augmentation techniques including simple operations such as cropping or flipping, just resizing all images to 224 × 224 for training and testing. Furthermore, the three attention modules including SE, CBAM, and SimAM all involve hyperparameter, and we apply the default settings suggested by the authors. Our HFAM does have any hyperparameter which is more convenient to employ. We implement each experiment five times independently to mitigate the effects of randomness and report the result in the form of "average ± deviation."

All experiments are trained under the same training parameter settings. The optimizer for training is SGD, and the batch size is 16. The detailed parameter settings for training are illustrated as follows. The learning rate for the last layer is set to 0.01 to converge rapidly and 0.001 for others. After every ten epochs, the learning rate is multiplied by 0.8. The weight decay parameter and the momentum in the training phase are set to 0.0005 and 0.9, respectively. The number of training epochs is set as 60. All experiments are conducted on a workstation with an NVIDIA GeForce RTX 3090 24-GB GPU.

C. Evaluation Metrics

Overall accuracy (OA) and confusion matrix (CM) are used to measure the classification results of different baseline and attention modules, which are widely used as metrics for image scene classification. The OA refers to the ratio that the number of correctly classified images is divided by the total number of images and reflects the performance of classification methods intuitively. As an informative table, the CM intuitively shows the error distribution of each scene classes. Moreover, the number of parameters of models and floating point operations (FLOPs) are two other important

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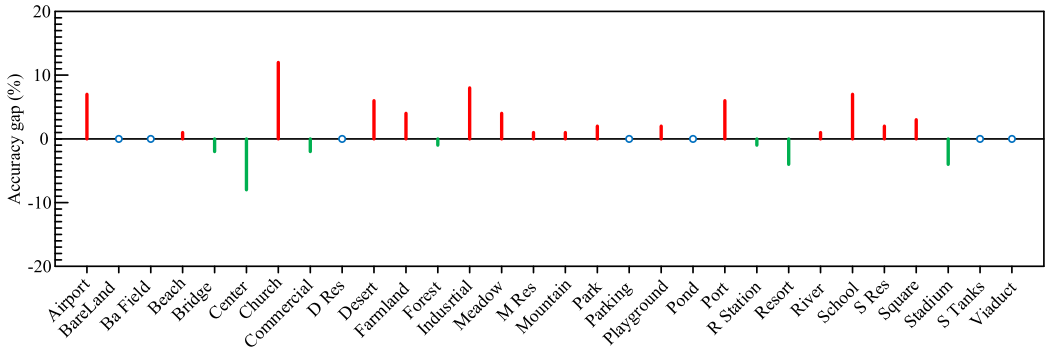


Fig. 8. Accuracy gaps of each class between baseline ResNet50 and network with our HFAM on the AID dataset. The length of red and green line segments, respectively, represent the improvement and reduction of our HFAM over the baseline network. Blue circles indicate that they get equal accuracy on this class.

metrics to measure model size and operational performance. In addition, whether hyperparameters are involved is also an important measure for ease of implementation.

D. Results Analysis

We compare our HFAM with the baseline CNN models and three popular attention modules: SE, CBAM, and SimAM. The performance of different attention modules is evaluated by three indicators: OA, number of parameters of models, and FLOPs. Furthermore, the CM is used to visualize the improvement of our HFAM over the baseline network for AID and NWPU datasets.

1) Result and Comparison on UCM: Table II offers the results of several indicators gained by different baseline and attention modules. Based on it, we made the following observations.

1) Under two different training ratios of 50% and 80%, our proposed HFAM observably improves the performance of the three different baseline models. Specifically, for a 50% training ratio, the insertion of HFAM resulted in enhancements of OA by 0.65%, 0.58%, and 0.72%, respectively for Vgg16, ResNet50, and ResNet101. When the training ratio is set as 80%, the improvements brought by HFAM are 0.43%, 0.29%, and 0.29% for three baselines.

2) Generally speaking, attention modules can improve the performance of the baseline networks on the UCM dataset except CBAM plugged into ResNet50 for both training ratios and SE inserted into ResNet101 for 80% training ratio. It is easy to find that compared with other attention modules, our proposed HFAM performs the best and obtains the highest OA in all cases.

3) Like SimAM, our HFAM does not add additional parameters to the baseline networks. The speed reflected by FLOPs also occupies a position second only to SimAM. It is also worth mentioning that HFAM is an attention module without any hyperparameters, which is a huge advantage and can be easily applied in various networks.

2) Result and Comparison on AID: To further prove the effectiveness of our HFAM, numerous experiments on AID dataset were conducted and the results are shown in Table III. It can be observed that as follows.

1) Consistent with the performance in UCM, our HFAM helps different baseline networks improve OA by 0.83%, 0.95%, and 1.38% for Vgg16, ResNet50, and ResNet101 under 20% training ratio. For an 80% training ratio, the advancements are 0.40%, 0.35%, and 0.24%, respectively. The insertion of HFAM significantly enhances the classification ability of the network. It can observe that as the training ratio increases, the improvement by HFAM on OA becomes lower, indicating that our HFAM improves the network more significantly in the case of fewer samples, which is also consistent with the performance in the UCM dataset.

2) Our proposed HFAM outperforms other used attention modules, achieving the highest or tie for the highest OA among all backbone networks under two different training ratios. In some cases, the utilization of certain attention modules can even damage the performance of the networks, such as the SimAM inserted into Vgg16 for 20% training ratio and three baselines for a 50% training ratio, and CBAM used in ResNet 50 and ResNet101 for 50% training ratio.

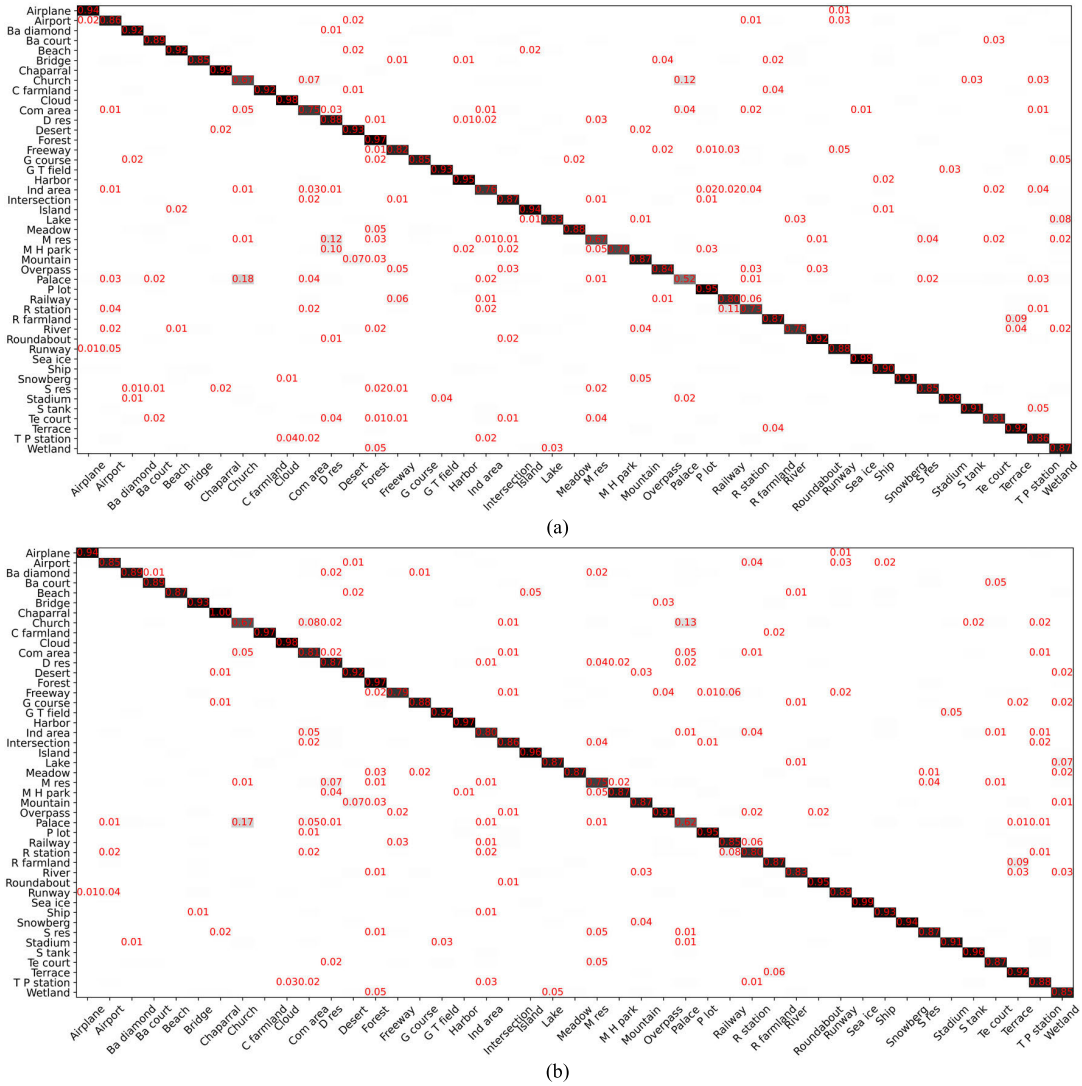
3) Two indicators of parameter volume and FLOPs further show the superiority of HFAM, especially when the performance of SimAM is so poor that it is not suitable for the application of remote-sensing image classification. In addition, the absence of hyperparameters in HFAM is a unique and superior advantage.

To better show the improvement by our proposed attention module, the confusion matrixes of ResNet50 with and without our proposed HFAM under a 20% training ratio are also presented in Fig. 7. In addition, we calculate the accuracy of each class, respectively, and compute the accuracy gap by subtracting the former from the latter and show the results in Fig. 8. It can be noted that our HFAM visibly improves the baseline on 16 categories including airport, beach, church, desert, farmland, industrial, meadow, medium residential, mountain, park, playground, port, river, school, sparse residential, and square. For categories whose semantic labels are mainly determined by key features, the improvement is quite remarkable: 7%, 12%, 8%, 6%, and 7% for airport, church, industrial, port, and school, respectively. Especially dealing with a complex background, the HFAM shows great advantage. For example, the baseline network misclassifies 4%, 5%, 1%, 2%, 3%, 2%, and 2% samples of the church as center, commercial, medium residential, resort, school, square, and storage tanks, while our HFAM only misrecognize 2%, 2%, and 2% of the church as center, commercial, and school. The range and value of misclassifications are reduced.

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TABLE IV
COMPARISON BETWEEN DIFFERENT ATTENTION MODULES ON NWPU DATASET (TRAINING RATIO = 10%)

Model	OA (%)	Parameters (M)	Hyperparameter	FLOPs (G)
Vgg16	86.44 ± 0.41	138.4110	×	248.3920
+ SE	86.65 ± 0.26	+ 0.0758	√	+ 0.0466
+ CBAM	86.84 ± 0.24	+ 0.0762	√	+ 0.0737
+ SimAM	85.83 ± 0.47	+ 0	√	+ 0
+ HFAM	**87.16 ± 0.16**	+ 0	×	+ 0.0197
ResNet50	87.43 ± 0.29	25.6021	×	66.1390
+ SE	89.09 ± 0.14	+ 0.6963	√	+ 0.0353
+ CBAM	88.11 ± 0.39	+ 0.6967	√	+ 0.0530
+ SimAM	87.83 ± 0.16	+ 0	√	+ 0
+ HFAM	**89.16 ± 0.06**	+ 0	×	+ 0.0241
ResNet101	87.97 ± 0.44	44.5942	×	125.8621
+ SE	89.39 ± 0.14	+ 0.6963	√	+ 0.0353
+ CBAM	88.33 ± 0.26	+ 0.6967	√	+ 0.0530
+ SimAM	88.64 ± 0.08	+ 0	√	+ 0
+ HFAM	**89.53 ± 0.29**	+ 0	×	+ 0.0241



(a)

(b)

Fig. 9. Comparison of CM produced by baseline (ResNet50) and the network inserted our proposed attention module on NWPU dataset with 10% training ratio: (a) ResNet50 and (b) ResNet50 + HFAM.

On the other hand, for categories whose semantic labels are mainly determined by color or texture features such as beach, desert, farmland, and meadow, our HFAM increase the accuracy by 1%, 6%, 4%, and 4%, respectively, which are also

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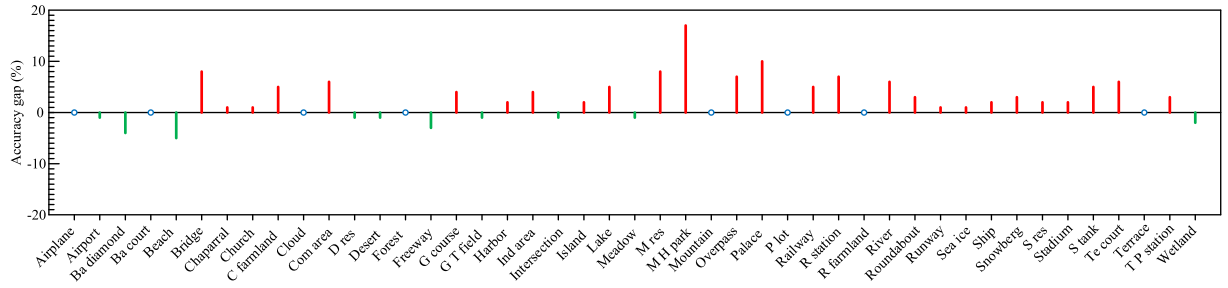


Fig. 10. Accuracy gaps of each class between baseline ResNet50 and network with our HFAM on NWPU dataset. The length of red and green line segments, respectively, represent the improvement and reduction of our HFAM over the baseline network. Blue circles indicate that they get equal accuracy on this class.

TABLE V

COMPARISON BETWEEN DIFFERENT STRUCTURES OF THE PROPOSED ATTENTION MODULES APPLIED AT DIFFERENT LOCATIONS ON UCM DATASET (TRAINING RATIO = 50%)

Structure	**Location of inserted attention module (HFAM)**				**OA (%)**
	Stage 1	**Stage 2**	**Stage 3**	**Stage 4**	
Backbone = Vgg16					95.77 ± 0.25
a	√				96.15 ± 0.37
b		√			96.09 ± 0.67
c			√		96.25 ± 0.68
d				√	96.36 ± 0.17
c + d			√	√	95.93 ± 0.37
b + c + d		√	√	√	95.92 ± 0.32
a + b + c + d (ours)	√	√	√	√	**96.42 ± 0.20**
Backbone = ResNet50					97.06 ± 0.22
a	√				96.72 ± 0.42
b		√			97.41 ± 0.20
c			√		96.93 ± 0.22
d				√	97.14 ± 0.32
c + d			√	√	97.24 ± 0.18
b + c + d		√	√	√	97.18 ± 0.16
a + b + c + d (ours)	√	√	√	√	**97.64 ± 0.18**
Backbone = ResNet101					96.74 ± 0.24
a	√				97.43 ± 0.31
b		√			97.16 ± 0.18
c			√		97.28 ± 0.36
d				√	97.37 ± 0.25
c + d			√	√	97.29 ± 0.39
b + c + d		√	√	√	96.90 ± 0.29
a + b + c + d (ours)	√	√	√	√	**97.46 ± 0.31**

considerable. Specifically, the baseline network misclassifies desert as bareland and meadow as forest, due to the similar uniform texture features that cannot be well recognized by it. It is obvious that our HFAM can distinguish different texture features by quantifying the difference of pixels in the feature map.

The reduction in accuracy by our HFAM occurs in these seven categories: bridge, center, commercial, forest, railway station, resort, and stadium with 2%, 8%, 2%, 1%, 1%, 4%, and 4% reduction, respectively. The main misclassification occurred in three categories including center (8%), resort (4%), and stadium (4%). Specifically, the serious misclassification phenomenon of the center is mainly due to its similar circular features of key object of church, industrial, square, and stadium. For resort, the main primary cause of decline is the increased confusion with airport, school, and square. Similarly, the reduction in the performance of the stadium is due to increased confusion with center and playground. Overall, our proposed HFAM improves the accuracy of the baseline network in most cases and obtains outstanding results.

3) Result and Comparison on NWPU: In addition to the experiments on the former two datasets, to further verify the effectiveness of HFAM, the NWPU dataset with larger scale samples was utilized. Table IV shows the results and several observations can be caught.

1) The implementation of the proposed HFAM contributes to increases on OA by 0.72%, 1.73%, and 1.56% for Vgg16, ResNet50, and ResNet101 separately with a 10% training ratio. This improvement is quite remarkable especially for the two ResNet networks, considering that any data augmentation techniques are not adapted in all experiments and the training ratio is set to a small number 10%. We can reasonably infer that the HFAM can significantly improve the RSSC ability of networks in the case of small samples.

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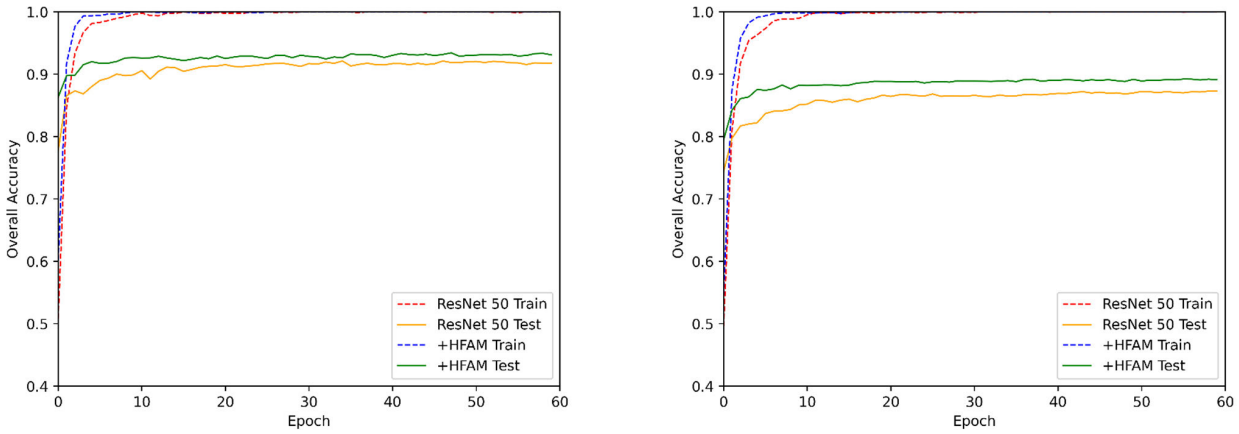


Fig. 11. Comparisons of training and test curves for ResNet50 with and without our proposed HFAM on AID with 20% training ratio and NWPU datasets with 10% training ratio, respectively. It can be seen that our HFAM improves the performance and converges faster compared with the baseline model: (a) AID dataset and (b) NWPU dataset.

2) For the more challenging large-scale NWPU dataset, our HFAM performs better than other plug-and-play attention modules on the three baseline networks, which shows its high robustness and high suitability for RSSC.

3) Neither HFAM nor SimAM increases the parameter quantity of the networks compared with other plug-and-play attention modules. It can be seen that the SimAM shows the lowest FLOPs and HFAM gets the second lowest place. On the other hand, HFAM has no hyperparameters, which avoids extensive experimentation to determine. Considering the indifferent performance of SimAM that even lowered the OA of baseline VGG16, our HFAM is undoubtedly the most suitable plug-and-play attention module for RSSC among used attention modules.

Same as experiments in the AID dataset, to further demonstrate the improvement of our HFAM over the baseline network ResNet50 on the NWPU dataset, the confusion matrixes and accuracy gap of each class are shown in Figs. 9 and 10, respectively. It can be clearly observed that our proposed HFAM significantly improves the performance of the baseline network in most cases. Specifically, it brings accuracy improvements for the baseline network on 27 classes and improved by more than 2% on 18 classes of them. These categories and their enhancements are: bridge (8%), circular farmland (5%), commercial area (6%), golf course (4%), industrial area (4%), lake (5%), medium residential (8%), mobile home park (17%), overpass (7%), palace (10%), railway (5%), railway station (7%), river (6%), roundabout (3%), snowberg (3%), storage tank (5%), tennis court (6%), and thermal power station (3%). At the same time, the HFAM reduces the accuracy on ten classes compared with the baseline network. However, these reductions are mostly tiny, with only three classes exceeding 2%. They are baseball diamond, beach, and freeway and their reductions are 4%, 5%, and 3%, respectively. Overall, it is obvious that the improvement of our HFAM over the baseline network is very significant.

E. Ablation Study

To assess the influence of the location of our HFAM inserted into networks, we implement a series of experiments on the UCM dataset with 50% training ratio. We consider six possible insertion cases: (a) after stage1, (b) after stage 2, (c) after stage 3, (d) after stage 4, (c + d) after stage 3 and stage 4, and (b + c + d) after stage 2, stage 3, and stage4. For Vgg16, we correspond its fourth, seventh, tenth, and last convolutional layers to stage 2, stage 3, stage 4, and stage 5 of the ResNet structure. The results are reported in Table V. It is obvious that the performance is best when the HFAM is introduced in each stage for the three different backbones. Since the insertion of HFAM does not add any additional parameters and the increase in the amount of calculation is also very low, it is more recommended to be used after all stages.

TABLE VI
COMPARISON WITH STATE-OF-THE-ART METHODS AT OA ON THE UCM DATASET

Method	Training Ratio		HP[1]
	50%	80%	
GBNet	97.07 ± 0.19	98.57 ± 0.48	√
ResNet50 + EAM	-	98.98 ± 0.37	√
MG-CAP	-	99.00 ± 0.10	√
ResNet50 + DOCP	-	99.15 ± 0.40	√
ACR-MLFF	97.99 ± 0.26	99.37 ± 0.15	√
ResNet50 without DA[2]	97.06 ± 0.22	98.33 ± 0.34	×
ResNet50 with DA	98.17 ± 0.08	99.19 ± 0.23	×
ResNet50 + HFAM (ours) without DA	97.64 ± 0.18	98.62 ± 0.11	×
ResNet50 + HFAM (ours) with DA	**98.35 ± 0.14**	**99.43 ± 0.13**	×

HP: Hyperparameter
DA: Data Augmentation

F. Efficiency and Visual Comparison

In order to more intuitively show the rapidity and effectiveness of our HFAM, we display the training and test curves of ResNet50 with HFAM on AID under 20% ratio training and NWPU datasets, respectively, in Fig. 11. It can be found that our proposed HFAM can improve the accuracy of baseline network on test dataset and converge faster than baseline model during training.

Furthermore, in order to better demonstrate the effect of different attention modules, for different models trained on the AID dataset with a 20% training ratio, we apply

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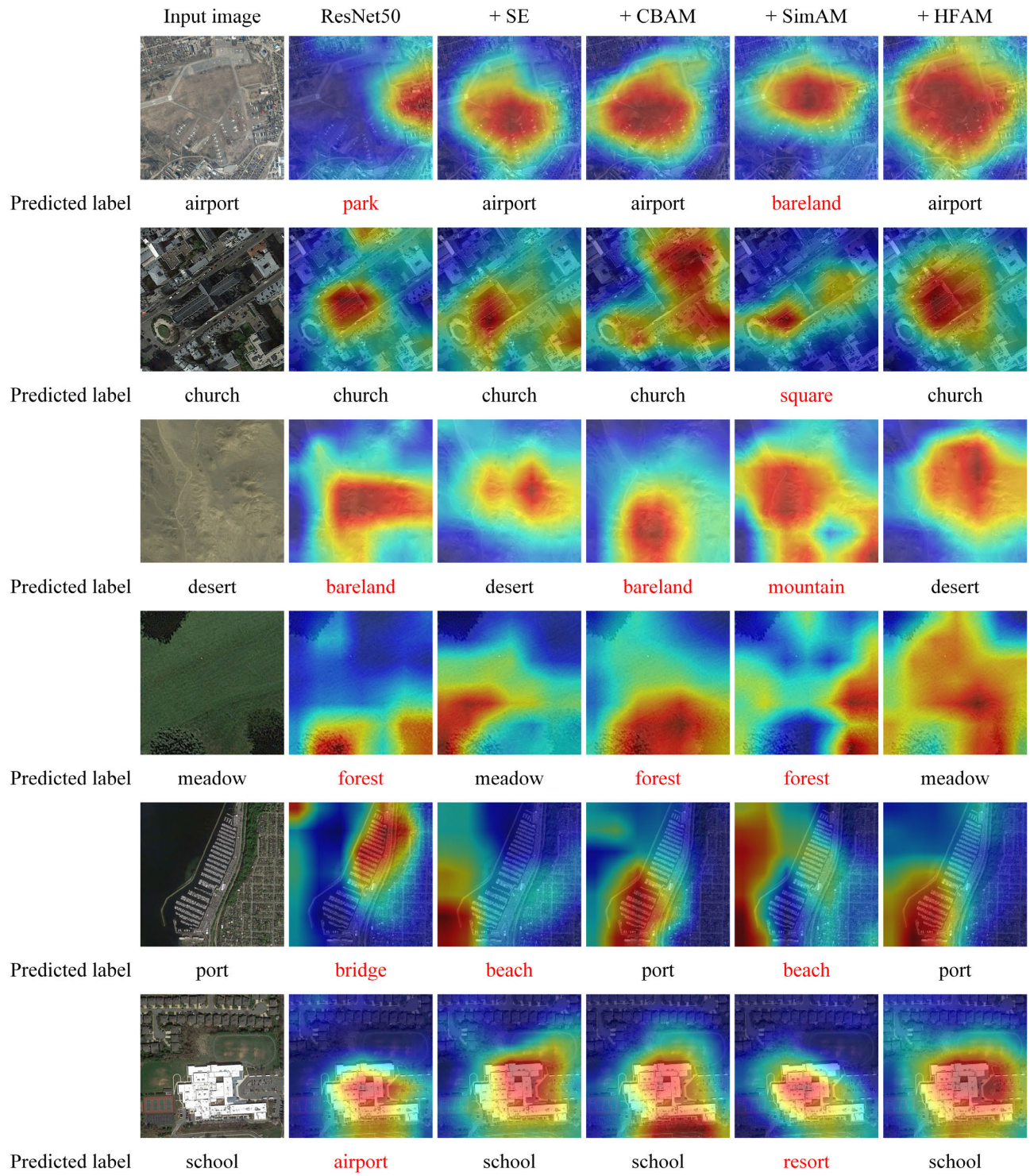


Fig. 12. Visualization of different attention modules integrated with ResNet50 on the AID dataset by Grad-CAM [48]. We generate the visualization results using the last convolutional outputs of networks. The real label and predicted labels by different models of each input image are displayed beneath corresponding images.

the Grad-CAM [48] on images from the AID test set. As illustrated in Fig. 12, it can be observed that as follows.

1) Compared with baseline network and other attention modules, our proposed HFAM can obtain refined features focusing on key and main objects when dealing with complex scenes such as airport, church, and school. The heatmaps generated by Grad-CAM with HFAM are more accurate in location than all other models that predicted correctly.

2) Our HFAM can effectively identify remote-sensing scenes whose semantic labels are mainly determined by large-scale texture, color, and other characteristic. For example, the sample of the desert is often misclassified as bareland and mountain because of similar texture features and the sample of

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TABLE VII
COMPARISON WITH STATE-OF-THE-ART METHODS AT OA ON THE AID DATASET

Method	Training Ratio		HP[1]
	20%	50%	
GBNet	92.20 ± 0.23	95.48 ± 0.12	√
ResNet50 + EAM	93.64 ± 0.25	96.62 ± 0.13	√
MG-CAP	93.34 ± 0.18	96.12 ± 0.12	√
ResNet50 + DOCP	-	-	√
ACR-MLFF	92.73 ± 0.12	95.06 ± 0.33	√
ResNet50 without DA[2]	92.16 ± 0.18	95.51 ± 0.15	×
ResNet50 with DA	93.63 ± 0.14	96.29 ± 0.09	×
ResNet50 + HFAM (ours) without DA	93.11 ± 0.20	95.86 ± 0.15	×
ResNet50 + HFAM (ours) with DA	**94.71 ± 0.06**	**96.73 ± 0.04**	×

HP: Hyperparameter
DA: Data Augmentation

TABLE VIII
COMPARISON WITH STATE-OF-THE-ART METHODS AT OA ON THE NWPU DATASET

Method	Training Ratio = 10	HP[1]
GBNet	-	√
ResNet50 + EAM	**90.87 ± 0.15**	√
MG-CAP	90.83 ± 0.12	√
ResNet50 + DOCP	89.74 ± 0.21	√
ACR-MLFF	90.01 ± 0.33	√
ResNet50 without DA[2]	87.43 ± 0.29	×
ResNet50 with DA	90.03 ± 0.07	×
ResNet50 + HFAM (ours) without DA	89.16 ± 0.06	×
ResNet50 + HFAM (ours) with DA	90.81 ± 0.11	×

HP: Hyperparameter
DA: Data Augmentation

the meadow is misidentified as forest due to the interference of trees in image by baseline and most attention modules, while our HFAM can excellently handle this type of remote-sensing imagery.

G. Comparison With State-Of-The-Art Methods

In this section, we compare our HFAM on the three datasets with some state-of-the-art methods proposed since 2020: GBNet [51], MG-CAP [52], ResNet50 + EAM [35], ResNet50 + DOCP [53], and ACR-MLFF [54]. We select ResNet50 as the backbone because it is a widely used CNN baseline architecture for most RSSC methods. Data augmentation is a commonly used trick in the task of image classification and all these previous methods for RSSC have adopted data augmentation technology. In order to get a fair comparison, we reperform our experiments with simple data augmentations including random horizontal flipping and random cropping while other experimental settings remain unchanged. The classification performance comparisons on three datasets are represented in Tables VI–VIII. It is obvious that our HFAM achieves quite competitive performance against these state-of-the-art methods without increasing the parameter amount of baseline model and introducing any hyperparameters.

VI. CONCLUSION

In this article, we proposed a lightweight, effective, and hyperparameter-free plug-and-play attention module named HFAM, which is designed from the characteristics of remote-sensing images and can adaptively generate attention weights based on feature map mathematical calculation. It does not involve complex operations including convolution and fully connection and has no hyperparameters. Hence, it can be expediently inserted into most existing CNN networks and trained by an end-to-end manner without increasing the number of parameters of the model.

Experiments based on three different baseline CNN models were conducted on three widely used open benchmark datasets. Without any data augmentation, our proposed HFAM outperform the baseline models and other attention modules. It simultaneously improved the performance of baseline models on two common types of remote-sensing scenes. Moreover, the training and test curves also showed that our HFAM can significantly improve the OA and converge faster than the baseline model during training. Meanwhile, the attention heatmaps generated by Grad-CAM visually displayed the superiority of our HFAM. Compared with state-of-the-art methods, it also shows considerable competitiveness.

In the future, we shall continue to explore and design simple and efficient attention modules for other remote-sensing tasks such as object detection and semantic segmentation. In addition, we plan to design similar simple but effective modules for transformer structure in the future study.

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